Chinese_topic_Exp.2: Statistical Report

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2019-06-17

Chinese topic processing

- Mandarin: 71id

Load relevant packages

```
library(plyr)
library(ggplot2)
library(reshape2)
library(scales)
library(MASS)
library(lme4)
```

```
## Loading required package: Matrix
```

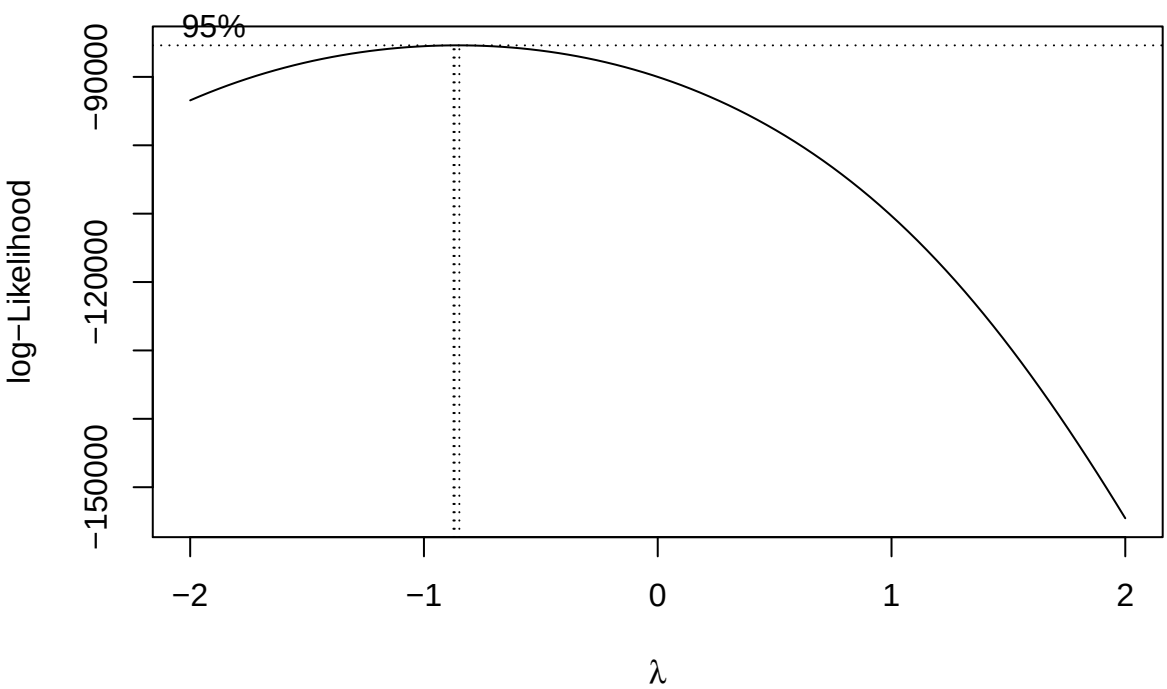
```
library(car)
library(languageR)
```

Import dataset: exp2. without context

```
topic2 <- read.csv("topic2_noContext_71id.csv")
```

```
topic2$subj <- as.factor(topic2$subj)
topic2$item <- as.factor(topic2$item)
```

```
boxcox <- boxcox(topic2$RT~ topic2$condition * topic2$subj)
```



```
##lamda peaks around -1, thus use rrt

topic2 <- subset(topic2, topic2$RT > 50 & topic2$ RT < 5000)
## eliminated 74 (0.36%) datapoints

topic2$rrt <- -1000/topic2$RT
```

sum-contrast coding

```
order <- ifelse(topic2$condition%in%c("a","c"),-1,1)
complexity <- ifelse(topic2$condition%in%c("a","b"),-1,1)
orderxcomplexity <- ifelse(topic2$condition%in%c("b","c"),-1,1)

topic2$order <- order
topic2$complexity <- complexity
topic2$orderxcomplexity <- orderxcomplexity
head(topic2, n=5)
```

```
##   subj item condition order complexity constituent position Q   RT
## 1    1    1         a    -1         -1         DCL        0 -  367
## 2    1    5         a    -1         -1         DCL        0 - 1029
## 3    1    9         a    -1         -1         DCL        0 -  351
## 4    1   13         a    -1         -1         DCL        0 -  341
## 5    1   17         a    -1         -1         DCL        0 -  370
##          rrt orderxcomplexity
## 1 -2.7247956                1
## 2 -0.9718173                1
## 3 -2.8490028                1
## 4 -2.9325513                1
## 5 -2.7027027                1
```

analyses by word positions

pos4 (RC-V2): W5

```
m0.pos4.RCV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1 + order + complexity + orderxcomplexity||subj)
              + (1 + order + complexity + orderxcomplexity||item),
              subset(topic2,position =="4"))
summary(m0.pos4.RCV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "4")
##
## REML criterion at convergence: 4512.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -8.3942 -0.5383 -0.0037  0.5459  3.5027
##
```

```
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.6239446 0.78990
##  subj.1   order            0.0000000 0.00000
##  subj.2   complexity       0.0005255 0.02292
##  subj.3   orderxcomplexity 0.0039304 0.06269
##  item     (Intercept)      0.0061296 0.07829
##  item.1   order            0.0014232 0.03773
##  item.2   complexity       0.0051388 0.07169
##  item.3   orderxcomplexity 0.0059465 0.07711
##  Residual                  0.7013086 0.83744
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.65770    0.09724 -27.332
## order            -0.12931    0.02170  -5.958
## complexity       -0.04454    0.02516  -1.770
## orderxcomplexity -0.05182    0.02674  -1.938
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```r
m1.pos4.RCV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
               + (1 + complexity+ orderxcomplexity ||subj)
               + (1 + order +complexity + orderxcomplexity||item),
               subset(topic2,position =="4"))  ## optimal model
summary(m1.pos4.RCV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     complexity | subj) + (0 + orderxcomplexity | subj)) + ((1 |
##     item) + (0 + order | item) + (0 + complexity | item) + (0 +
##     orderxcomplexity | item))
##    Data: subset(topic2, position == "4")
##
## REML criterion at convergence: 4512.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -8.3942 -0.5383 -0.0037  0.5459  3.5027
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.6239449 0.78990
##  subj.1   complexity       0.0005255 0.02292
##  subj.2   orderxcomplexity 0.0039304 0.06269
##  item     (Intercept)      0.0061296 0.07829
##  item.1   order            0.0014232 0.03773
##  item.2   complexity       0.0051388 0.07169
##  item.3   orderxcomplexity 0.0059465 0.07711
##  Residual                  0.7013086 0.83744
```

```
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.65770    0.09724 -27.332
## order            -0.12931    0.02170  -5.958
## complexity       -0.04454    0.02516  -1.770
## orderxcomplexity -0.05182    0.02674  -1.938
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

pos4.DE: W6

```
m0.pos5.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
          + (1+ order + complexity + orderxcomplexity||subj)
          + (1 + order + complexity + orderxcomplexity||item),
                   subset(topic2,position =="5"))
summary(m0.pos5.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "5")
##
## REML criterion at convergence: 4143.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.7246 -0.5448 -0.0351  0.5335  4.3963
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.41927  0.64751
##  subj.1   order            0.01206  0.10983
##  subj.2   complexity       0.00000  0.00000
##  subj.3   orderxcomplexity 0.00000  0.00000
##  item     (Intercept)      0.00516  0.07184
##  item.1   order            0.00000  0.00000
##  item.2   complexity       0.00000  0.00000
##  item.3   orderxcomplexity 0.00443  0.06656
##  Residual                  0.56656  0.75270
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.71904    0.08033  -33.85
## order            -0.16102    0.02242   -7.18
## complexity       -0.04176    0.01824   -2.29
```

```
## orderxcomplexity -0.03655    0.02274   -1.61
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.pos5.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
                   + (1+ order ||subj)
                   + (1 + orderxcomplexity||item),
                   subset(topic2,position =="5")) ## optimal model
summary(m1.pos5.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj)) + ((1 | item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "5")
##
## REML criterion at convergence: 4143.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.7246 -0.5448 -0.0351  0.5335  4.3963
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.41927  0.64751
##  subj.1   order            0.01206  0.10983
##  item     (Intercept)      0.00516  0.07184
##  item.1   orderxcomplexity 0.00443  0.06656
##  Residual                  0.56656  0.75270
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.71904    0.08033  -33.85
## order            -0.16102    0.02241   -7.18
## complexity       -0.04176    0.01824   -2.29
## orderxcomplexity -0.03655    0.02274   -1.61
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

pos6.HN: W7

```
m0.pos6.HN <- lmer(rrt ~ order + complexity + orderxcomplexity
           + (1+ order + complexity + orderxcomplexity||subj)
           + (1 + order + complexity + orderxcomplexity||item),
          subset(topic2,position =="6"))
summary(m0.pos6.HN)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "6")
##
## REML criterion at convergence: 4023.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -9.8713 -0.5842 -0.0531  0.5535  3.7649
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.4983261 0.70592
##  subj.1   order            0.0004601 0.02145
##  subj.2   complexity       0.0031902 0.05648
##  subj.3   orderxcomplexity 0.0000000 0.00000
##  item     (Intercept)      0.0029835 0.05462
##  item.1   order            0.0013869 0.03724
##  item.2   complexity       0.0000000 0.00000
##  item.3   orderxcomplexity 0.0006728 0.02594
##  Residual                  0.5318920 0.72931
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.845470   0.086344  -32.95
## order            -0.155116   0.019407   -7.99
## complexity       -0.019335   0.018902   -1.02
## orderxcomplexity -0.007546   0.018450   -0.41
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.001
## ordrxcmplxt  0.000 -0.001 -0.001
```

```
m1.pos6.HN <- lmer(rrt ~ order + complexity + orderxcomplexity
          + (1+ order + complexity ||subj)
          + (1 + order + orderxcomplexity||item),
          subset(topic2,position =="6"))  ##optimal model
summary(m1.pos6.HN)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj)) + ((1 | item) +
##     (0 + order | item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "6")
##
## REML criterion at convergence: 4023.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -9.8713 -0.5842 -0.0531  0.5535  3.7649
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.4983260 0.70592
##  subj.1   order            0.0004601 0.02145
##  subj.2   complexity       0.0031902 0.05648
##  item     (Intercept)      0.0029835 0.05462
##  item.1   order            0.0013869 0.03724
##  item.2   orderxcomplexity 0.0006728 0.02594
##  Residual                  0.5318919 0.72931
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.845470   0.086344  -32.95
## order            -0.155116   0.019407   -7.99
## complexity       -0.019335   0.018902   -1.02
## orderxcomplexity -0.007546   0.018450   -0.41
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.001
## ordrxcmplxt  0.000 -0.001 -0.001
```

pos7.ADV:W8

```r
m0.pos7.ADV <- lmer(rrt ~ order + complexity + orderxcomplexity
            + (1+ order + complexity + orderxcomplexity||subj)
            + (1 + order + complexity + orderxcomplexity||item),
            subset(topic2,position =="7"))
summary(m0.pos7.ADV)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "7")
##
## REML criterion at convergence: 4188.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.0549 -0.5695 -0.0035  0.5445  3.9090
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.537620 0.73323
##  subj.1   order            0.003580 0.05984
##  subj.2   complexity       0.000000 0.00000
##  subj.3   orderxcomplexity 0.000000 0.00000
##  item     (Intercept)      0.001726 0.04155
##  item.1   order            0.005279 0.07266
```

```
##  item.2   complexity       0.010495 0.10245
##  item.3   orderxcomplexity 0.002503 0.05003
##  Residual                  0.576819 0.75949
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.78744    0.08935 -31.198
## order            -0.12133    0.02468  -4.917
## complexity       -0.04861    0.02785  -1.745
## orderxcomplexity -0.01860    0.02104  -0.884
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```r
m1.pos7.ADV <- lmer(rrt ~ order + complexity + orderxcomplexity
           + (1+ order ||subj)
           + (1 + order + complexity + orderxcomplexity||item),
           subset(topic2,position =="7")) ## optimal model
summary(m1.pos7.ADV)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj)) + ((1 | item) + (0 + order | item) + (0 +
##     complexity | item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "7")
##
## REML criterion at convergence: 4188.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.0549 -0.5695 -0.0035  0.5445  3.9090
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.537620 0.73323
##  subj.1   order            0.003580 0.05984
##  item     (Intercept)      0.001726 0.04155
##  item.1   order            0.005279 0.07266
##  item.2   complexity       0.010495 0.10245
##  item.3   orderxcomplexity 0.002503 0.05003
##  Residual                  0.576819 0.75949
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.78744    0.08935 -31.198
## order            -0.12133    0.02468  -4.917
## complexity       -0.04861    0.02785  -1.745
## orderxcomplexity -0.01860    0.02104  -0.884
##
## Correlation of Fixed Effects:
```

```
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

pos8.quantifier: W9

```
m0.pos8.quantifier <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1+ order + complexity + orderxcomplexity||subj)
              + (1 + order + complexity + orderxcomplexity||item),
             subset(topic2,position =="8"))
summary(m0.pos8.quantifier)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "8")
##
## REML criterion at convergence: 4143.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -8.6752 -0.4929 -0.0237  0.5229  3.8515
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.450169 0.67095
##  subj.1   order            0.010080 0.10040
##  subj.2   complexity       0.001947 0.04413
##  subj.3   orderxcomplexity 0.000000 0.00000
##  item     (Intercept)      0.005823 0.07631
##  item.1   order            0.000000 0.00000
##  item.2   complexity       0.004364 0.06606
##  item.3   orderxcomplexity 0.001604 0.04005
##  Residual                  0.562656 0.75010
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.89583    0.08315  -34.83
## order            -0.03872    0.02173   -1.78
## complexity        0.02732    0.02323    1.18
## orderxcomplexity  0.02131    0.01993    1.07
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m0.pos8.quantifier <- lmer(rrt ~ order + complexity + orderxcomplexity
                 + (1+ order + complexity ||subj)
                 + (1 + complexity + orderxcomplexity||item),
                 subset(topic2,position =="8")) ## optimal model
```

```r
summary(m0.pos8.quantifier)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj)) + ((1 | item) +
##     (0 + complexity | item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "8")
##
## REML criterion at convergence: 4143.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -8.6752 -0.4929 -0.0237  0.5229  3.8515
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.450169 0.67095
##  subj.1   order            0.010080 0.10040
##  subj.2   complexity       0.001947 0.04413
##  item     (Intercept)      0.005823 0.07631
##  item.1   complexity       0.004364 0.06606
##  item.2   orderxcomplexity 0.001604 0.04005
##  Residual                  0.562656 0.75010
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.89583    0.08315  -34.83
## order            -0.03872    0.02173   -1.78
## complexity        0.02732    0.02323    1.18
## orderxcomplexity  0.02131    0.01993    1.07
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

pos9: W10

```r
m0.pos9.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
           + (1+ order + complexity + orderxcomplexity||subj)
          + (1 + order + complexity + orderxcomplexity||item),
          subset(topic2,position =="9"))
summary(m0.pos9.MV1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "9")
##
## REML criterion at convergence: 3853.3
```

```
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -8.2018 -0.5713 -0.0183  0.5724  4.3271
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      4.023e-01 6.343e-01
##  subj.1   order            1.590e-17 3.988e-09
##  subj.2   complexity       3.986e-03 6.313e-02
##  subj.3   orderxcomplexity 0.000e+00 0.000e+00
##  item     (Intercept)      0.000e+00 0.000e+00
##  item.1   order            0.000e+00 0.000e+00
##  item.2   complexity       2.942e-03 5.424e-02
##  item.3   orderxcomplexity 0.000e+00 0.000e+00
##  Residual                  4.844e-01 6.960e-01
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.775323   0.077139  -35.98
## order             0.015586   0.016867    0.92
## complexity        0.001792   0.021522    0.08
## orderxcomplexity  0.025726   0.016866    1.53
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.000
## ordrxcmplxt  0.000 -0.001  0.000
```

```
m1.pos9.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
                    + (1+  complexity ||subj)
                    + (1 + complexity ||item),
                    subset(topic2,position =="9")) ## optimal model
summary(m1.pos9.MV1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     complexity | subj)) + ((1 | item) + (0 + complexity | item))
##    Data: subset(topic2, position == "9")
##
## REML criterion at convergence: 3853.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -8.2018 -0.5713 -0.0183  0.5724  4.3271
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  subj     (Intercept) 4.023e-01 6.343e-01
##  subj.1   complexity  3.986e-03 6.313e-02
##  item     (Intercept) 1.101e-14 1.049e-07
##  item.1   complexity  2.942e-03 5.424e-02
##  Residual             4.844e-01 6.960e-01
```

```
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.775323   0.077139  -35.98
## order             0.015586   0.016867    0.92
## complexity        0.001792   0.021522    0.08
## orderxcomplexity  0.025726   0.016866    1.53
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.000
## ordrxcmplxt  0.000 -0.001  0.000
```

pos10: W11

```r
m0.pos10.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1+ order + complexity + orderxcomplexity||subj)
             + (1 + order + complexity + orderxcomplexity||item),
             subset(topic2,position =="10"))
summary(m0.pos10.MV1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "10")
##
## REML criterion at convergence: 3843.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.5571 -0.5534 -0.0260  0.6153  3.4254
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.313300 0.55973
##  subj.1   order            0.002349 0.04847
##  subj.2   complexity       0.000000 0.00000
##  subj.3   orderxcomplexity 0.000000 0.00000
##  item     (Intercept)      0.001807 0.04251
##  item.1   order            0.000000 0.00000
##  item.2   complexity       0.002290 0.04785
##  item.3   orderxcomplexity 0.001009 0.03177
##  Residual                  0.486646 0.69760
## Number of obs: 1702, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -1.93664    0.06909 -28.029
## order            -0.03383    0.01786  -1.894
## complexity       -0.04913    0.01953  -2.516
```

```
## orderxcomplexity -0.03137    0.01811  -1.732
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.000
## ordrxcmplxt  0.000  0.000 -0.001
```

```
m1.pos10.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
                     + (1+ order ||subj)
                     + (1 + complexity + orderxcomplexity||item),
                     subset(topic2,position =="10")) ## optimal model
summary(m1.pos10.MV1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj)) + ((1 | item) + (0 + complexity | item) +
##     (0 + orderxcomplexity | item))
##    Data: subset(topic2, position == "10")
##
## REML criterion at convergence: 3843.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.5571 -0.5534 -0.0260  0.6153  3.4254
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.313300 0.55973
##  subj.1   order            0.002349 0.04847
##  item     (Intercept)      0.001807 0.04251
##  item.1   complexity       0.002290 0.04785
##  item.2   orderxcomplexity 0.001009 0.03177
##  Residual                  0.486646 0.69760
## Number of obs: 1702, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -1.93664    0.06909 -28.029
## order            -0.03383    0.01786  -1.894
## complexity       -0.04913    0.01953  -2.516
## orderxcomplexity -0.03137    0.01811  -1.732
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.000
## ordrxcmplxt  0.000  0.000 -0.001
```

analyses by part of speech

RC-V2:

```
m0.RCV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
                + (1+ order + complexity + orderxcomplexity||subj)
```

```r
                + (1 + order + complexity + orderxcomplexity||item),
                subset(topic2,constituent =="RC-V2"))
summary(m0.RCV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, constituent == "RC-V2")
##
## REML criterion at convergence: 4373.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -9.0888 -0.5530 -0.0148  0.5702  4.2529
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.617279 0.78567
##  subj.1   order            0.008111 0.09006
##  subj.2   complexity       0.000000 0.00000
##  subj.3   orderxcomplexity 0.001614 0.04017
##  item     (Intercept)      0.007676 0.08761
##  item.1   order            0.000000 0.00000
##  item.2   complexity       0.006482 0.08051
##  item.3   orderxcomplexity 0.000000 0.00000
##  Residual                  0.642491 0.80156
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.599306   0.096907 -26.823
## order            -0.070919   0.022167  -3.199
## complexity       -0.006314   0.025439  -0.248
## orderxcomplexity -0.013643   0.019996  -0.682
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```r
m1.RCV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
                + (1+ order + orderxcomplexity||subj)
                + (1 + complexity ||item),
                subset(topic2,constituent =="RC-V2")) ## optimal model
summary(m1.RCV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + orderxcomplexity | subj)) + ((1 | item) +
##     (0 + complexity | item))
##    Data: subset(topic2, constituent == "RC-V2")
```

```
##
## REML criterion at convergence: 4373.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -9.0888 -0.5530 -0.0148  0.5702  4.2529
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.617279 0.78567
##  subj.1   order            0.008111 0.09006
##  subj.2   orderxcomplexity 0.001614 0.04017
##  item     (Intercept)      0.007676 0.08761
##  item.1   complexity       0.006482 0.08051
##  Residual                  0.642491 0.80156
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.599306   0.096907 -26.823
## order            -0.070919   0.022167  -3.199
## complexity       -0.006314   0.025439  -0.248
## orderxcomplexity -0.013643   0.019996  -0.682
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

DE:

```
m0.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1+ order + complexity + orderxcomplexity||subj)
              + (1 + order + complexity + orderxcomplexity||item),
              subset(topic2,constituent =="DE"))
summary(m0.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, constituent == "DE")
##
## REML criterion at convergence: 4179.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.6164 -0.5225 -0.0363  0.5012  4.2204
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.4193492 0.64757
```

```
##  subj.1   order            0.0089323 0.09451
##  subj.2   complexity       0.0000000 0.00000
##  subj.3   orderxcomplexity 0.0000000 0.00000
##  item     (Intercept)      0.0087598 0.09359
##  item.1   order            0.0007131 0.02670
##  item.2   complexity       0.0040746 0.06383
##  item.3   orderxcomplexity 0.0035336 0.05944
##  Residual                  0.5765317 0.75930
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.678975   0.081300  -32.95
## order            -0.120868   0.022225   -5.44
## complexity       -0.001839   0.022543   -0.08
## orderxcomplexity  0.003641   0.022037    0.17
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1+ order ||subj)
              + (1 + order + complexity + orderxcomplexity||item),
              subset(topic2,constituent =="DE")) ## optimal model
summary(m1.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj)) + ((1 | item) + (0 + order | item) + (0 +
##     complexity | item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, constituent == "DE")
##
## REML criterion at convergence: 4179.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.6164 -0.5225 -0.0363  0.5012  4.2204
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.4193492 0.64757
##  subj.1   order            0.0089323 0.09451
##  item     (Intercept)      0.0087598 0.09359
##  item.1   order            0.0007131 0.02670
##  item.2   complexity       0.0040746 0.06383
##  item.3   orderxcomplexity 0.0035336 0.05944
##  Residual                  0.5765316 0.75930
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.678975   0.081300  -32.95
```

```
## order            -0.120868   0.022225   -5.44
## complexity       -0.001839   0.022543   -0.08
## orderxcomplexity  0.003641   0.022037    0.17
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

HN:

```r
m0.HN <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1+ order + complexity + orderxcomplexity||subj)
              + (1 + order + complexity + orderxcomplexity||item),
              subset(topic2,constituent =="HN"))
summary(m0.HN)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, constituent == "HN")
##
## REML criterion at convergence: 3937.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.0531 -0.5941 -0.0515  0.5436  3.7299
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      4.789e-01 0.692037
##  subj.1   order            7.667e-05 0.008756
##  subj.2   complexity       9.159e-04 0.030263
##  subj.3   orderxcomplexity 0.000e+00 0.000000
##  item     (Intercept)      3.512e-03 0.059261
##  item.1   order            2.804e-03 0.052950
##  item.2   complexity       3.350e-04 0.018303
##  item.3   orderxcomplexity 0.000e+00 0.000000
##  Residual                  5.064e-01 0.711646
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.831728   0.084788  -33.40
## order            -0.141414   0.020379   -6.94
## complexity       -0.005659   0.018008   -0.31
## orderxcomplexity  0.006105   0.017246    0.35
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
```

```
## complexity   0.000  0.000
## ordrxcmplxt  0.000 -0.001 -0.001
```

```
m1.HN <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1+ order + complexity ||subj)
              + (1 + order + complexity + orderxcomplexity||item),
              subset(topic2,constituent =="HN"))
summary(m1.HN)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj)) + ((1 | item) +
##     (0 + order | item) + (0 + complexity | item) + (0 + orderxcomplexity |
##     item))
##    Data: subset(topic2, constituent == "HN")
##
## REML criterion at convergence: 3937.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.0531 -0.5941 -0.0515  0.5436  3.7299
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      4.789e-01 0.692037
##  subj.1   order            7.665e-05 0.008755
##  subj.2   complexity       9.159e-04 0.030263
##  item     (Intercept)      3.512e-03 0.059261
##  item.1   order            2.804e-03 0.052950
##  item.2   complexity       3.350e-04 0.018303
##  item.3   orderxcomplexity 0.000e+00 0.000000
##  Residual                  5.064e-01 0.711646
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.831728   0.084788  -33.40
## order            -0.141414   0.020379   -6.94
## complexity       -0.005659   0.018008   -0.31
## orderxcomplexity  0.006105   0.017246    0.35
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.000
## ordrxcmplxt  0.000 -0.001 -0.001
```

ADV:

```
m0.ADV <- lmer(rrt ~ order + complexity + orderxcomplexity
               + (1+ order + complexity + orderxcomplexity||subj)
               + (1 + order + complexity + orderxcomplexity||item),
               subset(topic2,constituent =="ADV"))
summary(m0.ADV)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, constituent == "ADV")
##
## REML criterion at convergence: 4184.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.8685 -0.5732 -0.0294  0.5419  3.9625
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.561307 0.74920
##  subj.1   order            0.006584 0.08114
##  subj.2   complexity       0.004707 0.06860
##  subj.3   orderxcomplexity 0.000000 0.00000
##  item     (Intercept)      0.003539 0.05949
##  item.1   order            0.003268 0.05716
##  item.2   complexity       0.008273 0.09095
##  item.3   orderxcomplexity 0.002378 0.04876
##  Residual                  0.568207 0.75380
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.764124   0.091579 -30.183
## order            -0.098089   0.023715  -4.136
## complexity       -0.025367   0.027286  -0.930
## orderxcomplexity  0.004611   0.020799   0.222
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```r
m1.ADV <- lmer(rrt ~ order + complexity + orderxcomplexity
               + (1+ order + complexity ||subj)
               + (1 + order + complexity + orderxcomplexity||item),
               subset(topic2,constituent =="ADV")) ## optimal model
summary(m1.ADV)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj)) + ((1 | item) +
##     (0 + order | item) + (0 + complexity | item) + (0 + orderxcomplexity |
##     item))
##    Data: subset(topic2, constituent == "ADV")
##
## REML criterion at convergence: 4184.6
##
## Scaled residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -6.8685 -0.5732 -0.0294  0.5419  3.9625
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.561308 0.74920
##  subj.1   order            0.006584 0.08114
##  subj.2   complexity       0.004707 0.06860
##  item     (Intercept)      0.003539 0.05949
##  item.1   order            0.003268 0.05716
##  item.2   complexity       0.008273 0.09095
##  item.3   orderxcomplexity 0.002378 0.04876
##  Residual                  0.568207 0.75380
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.764124   0.091579 -30.183
## order            -0.098089   0.023715  -4.136
## complexity       -0.025367   0.027286  -0.930
## orderxcomplexity  0.004611   0.020799   0.222
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

quantifier:

```r
m0.quantifier <- lmer(rrt ~ order + complexity + orderxcomplexity
               + (1+ order + complexity + orderxcomplexity||subj)
               + (1 + order + complexity + orderxcomplexity||item),
               subset(topic2,constituent =="quantifier"))
summary(m0.quantifier)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, constituent == "quantifier")
##
## REML criterion at convergence: 4060.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -9.1017 -0.5303 -0.0175  0.5287  3.7618
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.4312433 0.65669
##  subj.1   order            0.0065253 0.08078
##  subj.2   complexity       0.0008594 0.02932
```

```
##  subj.3   orderxcomplexity 0.0000000 0.00000
##  item     (Intercept)      0.0048557 0.06968
##  item.1   order            0.0003355 0.01832
##  item.2   complexity       0.0049456 0.07032
##  item.3   orderxcomplexity 0.0005637 0.02374
##  Residual                  0.5393848 0.73443
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.914065   0.081196  -35.89
## order            -0.056825   0.020554   -2.76
## complexity        0.009219   0.023124    0.40
## orderxcomplexity  0.003253   0.018441    0.18
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.quantifier <- lmer(rrt ~ order + complexity + orderxcomplexity
               + (1+ order + complexity ||subj)
               + (1 + order + complexity + orderxcomplexity||item),
              subset(topic2,constituent =="quantifier")) ## optimal model
summary(m1.quantifier)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj)) + ((1 | item) +
##     (0 + order | item) + (0 + complexity | item) + (0 + orderxcomplexity |
##     item))
##    Data: subset(topic2, constituent == "quantifier")
##
## REML criterion at convergence: 4060.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -9.1017 -0.5303 -0.0175  0.5287  3.7618
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.4312434 0.65669
##  subj.1   order            0.0065253 0.08078
##  subj.2   complexity       0.0008594 0.02932
##  item     (Intercept)      0.0048557 0.06968
##  item.1   order            0.0003355 0.01832
##  item.2   complexity       0.0049456 0.07032
##  item.3   orderxcomplexity 0.0005637 0.02374
##  Residual                  0.5393848 0.73443
## Number of obs: 1704, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.914065   0.081196  -35.89
```

```
## order            -0.056825   0.020554   -2.76
## complexity        0.009219   0.023124    0.40
## orderxcomplexity  0.003253   0.018441    0.18
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

main-V1 ## ME of order, ME of complexity, and interaction

```r
m0.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
            + (1+ order + complexity + orderxcomplexity||subj)
           + (1 + order + complexity + orderxcomplexity||item),
           subset(topic2,constituent =="main-V1"))
summary(m0.MV1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, constituent == "main-V1")
##
## REML criterion at convergence: 3965.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -10.4139  -0.5192   0.0098   0.5637   4.5493
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.442991 0.66558
##  subj.1   order            0.000000 0.00000
##  subj.2   complexity       0.006244 0.07902
##  subj.3   orderxcomplexity 0.000000 0.00000
##  item     (Intercept)      0.001242 0.03524
##  item.1   order            0.000000 0.00000
##  item.2   complexity       0.003737 0.06113
##  item.3   orderxcomplexity 0.000000 0.00000
##  Residual                  0.513748 0.71676
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.84922    0.08120  -35.09
## order            -0.05821    0.01737   -3.35
## complexity       -0.07206    0.02335   -3.09
## orderxcomplexity -0.04812    0.01737   -2.77
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
```

```
## complexity   0.000  0.000
## ordrxcmplxt  0.000 -0.001  0.000
```

```
m1.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
               + (1+ complexity ||subj)
               + (1 + complexity ||item),
               subset(topic2,constituent =="main-V1")) ## optimal model
summary(m1.MV1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     complexity | subj)) + ((1 | item) + (0 + complexity | item))
##    Data: subset(topic2, constituent == "main-V1")
##
## REML criterion at convergence: 3965.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -10.4139  -0.5192   0.0098   0.5637   4.5493
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  subj     (Intercept) 0.442991 0.66558
##  subj.1   complexity  0.006244 0.07902
##  item     (Intercept) 0.001242 0.03524
##  item.1   complexity  0.003737 0.06113
##  Residual             0.513748 0.71676
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.84922    0.08120  -35.09
## order            -0.05821    0.01737   -3.35
## complexity       -0.07206    0.02335   -3.09
## orderxcomplexity -0.04812    0.01737   -2.77
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.000
## ordrxcmplxt  0.000 -0.001  0.000
```

main-V2: ME of order, ME of complexity, interactions

```
m0.MV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
               + (1+ order + complexity + orderxcomplexity||subj)
               + (1 + order + complexity + orderxcomplexity||item),
               subset(topic2,constituent =="main-V2"))
summary(m0.MV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
```

```
##    Data: subset(topic2, constituent == "main-V2")
##
## REML criterion at convergence: 4066
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.7228 -0.5520 -0.0002  0.5882  3.2397
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.3334990 0.57749
##  subj.1   order            0.0143350 0.11973
##  subj.2   complexity       0.0074067 0.08606
##  subj.3   orderxcomplexity 0.0176445 0.13283
##  item     (Intercept)      0.0002886 0.01699
##  item.1   order            0.0032804 0.05727
##  item.2   complexity       0.0025847 0.05084
##  item.3   orderxcomplexity 0.0000000 0.00000
##  Residual                  0.5265266 0.72562
## Number of obs: 1702, groups:  subj, 71; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.16802    0.07084  -30.60
## order            -0.26521    0.02546  -10.42
## complexity       -0.28049    0.02283  -12.28
## orderxcomplexity -0.26252    0.02362  -11.11
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.000
## ordrxcmplxt  0.000  0.000 -0.001
```

```
m1.MV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
               + (1+ order + complexity + orderxcomplexity||subj)
               + (1 + order + complexity||item),
               subset(topic2,constituent =="main-V2")) ## optimal model
summary(m0.MV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic2, constituent == "main-V2")
##
## REML criterion at convergence: 4066
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.7228 -0.5520 -0.0002  0.5882  3.2397
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
```

```
##  subj     (Intercept)      0.3334990 0.57749
##  subj.1   order            0.0143350 0.11973
##  subj.2   complexity       0.0074067 0.08606
##  subj.3   orderxcomplexity 0.0176445 0.13283
##  item     (Intercept)      0.0002886 0.01699
##  item.1   order            0.0032804 0.05727
##  item.2   complexity       0.0025847 0.05084
##  item.3   orderxcomplexity 0.0000000 0.00000
##  Residual                  0.5265266 0.72562
## Number of obs: 1702, groups:  subj, 71; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -2.16802    0.07084  -30.60
## order            -0.26521    0.02546  -10.42
## complexity       -0.28049    0.02283  -12.28
## orderxcomplexity -0.26252    0.02362  -11.11
##
## Correlation of Fixed Effects:
##             (Intr) order  cmplxt
## order        0.000
## complexity   0.000  0.000
## ordrxcmplxt  0.000  0.000 -0.001
```

Unpack the interactions at MV1 & MV2

nested contrast

```
topic2$TSV_inMC <- ifelse(topic2$condition=="d",-1,
                          ifelse(topic2$condition=="c",1,0))
topic2$TSV_inRC <- ifelse(topic2$condition=="b",-1,
                          ifelse(topic2$condition=="a",1,0))

topic2$MC_inSVO <- ifelse(topic2$condition == "b",-1,
                          ifelse(topic2$condition == "d",1,0))
topic2$MC_inTSV <-ifelse(topic2$condition =="a",-1,
                         ifelse(topic2$condition == "c",1,0))
head(topic2, n=5)
```

```
##   subj item condition order complexity constituent position Q   RT
## 1    1    1         a    -1         -1         DCL        0 -  367
## 2    1    5         a    -1         -1         DCL        0 - 1029
## 3    1    9         a    -1         -1         DCL        0 -  351
## 4    1   13         a    -1         -1         DCL        0 -  341
## 5    1   17         a    -1         -1         DCL        0 -  370
##          rrt orderxcomplexity TSV_inMC TSV_inRC MC_inSVO MC_inTSV
## 1 -2.7247956                1        0        1        0       -1
## 2 -0.9718173                1        0        1        0       -1
## 3 -2.8490028                1        0        1        0       -1
## 4 -2.9325513                1        0        1        0       -1
## 5 -2.7027027                1        0        1        0       -1
```

nested comparision MV1:

```r
m0.MCV1.rrt.nested <-lmer(rrt ~ TSV_inMC + TSV_inRC + complexity
               + (1 + TSV_inMC + TSV_inRC + complexity||subj)
               + (1 + TSV_inMC + TSV_inRC + complexity||item),
             subset(topic2,constituent =="main-V1"), REML = FALSE)

summary(m0.MCV1.rrt.nested)  ###  t=4.33, TSV slower than SVO in MC
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula:
## rrt ~ TSV_inMC + TSV_inRC + complexity + ((1 | subj) + (0 + TSV_inMC |
##     subj) + (0 + TSV_inRC | subj) + (0 + complexity | subj)) +
##     ((1 | item) + (0 + TSV_inMC | item) + (0 + TSV_inRC | item) +
##         (0 + complexity | item))
##    Data: subset(topic2, constituent == "main-V1")
##
##      AIC      BIC   logLik deviance df.resid
##   3970.4   4041.1  -1972.2   3944.4     1690
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -10.4240  -0.5176   0.0099   0.5643   4.5490
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  subj     (Intercept) 4.365e-01 6.607e-01
##  subj.1   TSV_inMC    9.474e-15 9.733e-08
##  subj.2   TSV_inRC    0.000e+00 0.000e+00
##  subj.3   complexity  5.976e-03 7.730e-02
##  item     (Intercept) 1.226e-03 3.502e-02
##  item.1   TSV_inMC    0.000e+00 0.000e+00
##  item.2   TSV_inRC    0.000e+00 0.000e+00
##  item.3   complexity  3.348e-03 5.786e-02
##  Residual             5.133e-01 7.164e-01
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept) -2.84922    0.08063  -35.34
## TSV_inMC     0.10633    0.02455    4.33
## TSV_inRC     0.01011    0.02456    0.41
## complexity  -0.07206    0.02291   -3.14
##
## Correlation of Fixed Effects:
##            (Intr) TSV_MC TSV_RC
## TSV_inMC   0.000
## TSV_inRC   0.000  0.000
## complexity 0.000  0.000  0.001
```

```r
m0.MCV1.rrt.nested.2<-lmer(rrt~ MC_inSVO + MC_inTSV + order
              + (1 + MC_inSVO + MC_inTSV + order||subj)
             + (1 + MC_inSVO + MC_inTSV + order||item),
          subset(topic2,constituent =="main-V1"), REML = FALSE)

summary(m0.MCV1.rrt.nested.2) ##  t=-4.43, in SVO, MC faster than RC
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula:
## rrt ~ MC_inSVO + MC_inTSV + order + ((1 | subj) + (0 + MC_inSVO |
##     subj) + (0 + MC_inTSV | subj) + (0 + order | subj)) + ((1 |
##     item) + (0 + MC_inSVO | item) + (0 + MC_inTSV | item) + (0 +
##     order | item))
##    Data: subset(topic2, constituent == "main-V1")
##
##      AIC      BIC   logLik deviance df.resid
##   3972.8   4043.5  -1973.4   3946.8     1690
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -10.3583  -0.5249  -0.0008   0.5638   4.4473
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  subj     (Intercept) 4.369e-01 6.610e-01
##  subj.1   MC_inSVO    9.083e-03 9.530e-02
##  subj.2   MC_inTSV    0.000e+00 0.000e+00
##  subj.3   order       9.047e-16 3.008e-08
##  item     (Intercept) 1.202e-03 3.468e-02
##  item.1   MC_inSVO    0.000e+00 0.000e+00
##  item.2   MC_inTSV    2.285e-03 4.780e-02
##  item.3   order       0.000e+00 0.000e+00
##  Residual             5.170e-01 7.190e-01
## Number of obs: 1703, groups:  subj, 71; item, 24
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept) -2.84916    0.08067  -35.32
## MC_inSVO    -0.12006    0.02712   -4.43
## MC_inTSV    -0.02400    0.02649   -0.91
## order       -0.05823    0.01742   -3.34
##
## Correlation of Fixed Effects:
##          (Intr) MC_SVO MC_TSV
## MC_inSVO  0.000
## MC_inTSV  0.000  0.000
## order     0.000 -0.001  0.000
```

nested comparison # MCV2

```
m0.MCV2.rrt.nested <-lmer(rrt ~ TSV_inMC + TSV_inRC + complexity
             + (1 + TSV_inMC + TSV_inRC + complexity||subj)
            + (1 + TSV_inMC + TSV_inRC + complexity||item),
          subset(topic2,constituent =="main-V2"), REML = FALSE)

summary(m0.MCV2.rrt.nested)  ###  t=12.99, TSV slower than SVO in MC
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula:
## rrt ~ TSV_inMC + TSV_inRC + complexity + ((1 | subj) + (0 + TSV_inMC |
```

```
##     subj) + (0 + TSV_inRC | subj) + (0 + complexity | subj)) +
##     ((1 | item) + (0 + TSV_inMC | item) + (0 + TSV_inRC | item) +
##         (0 + complexity | item))
##    Data: subset(topic2, constituent == "main-V2")
##
##      AIC      BIC   logLik deviance df.resid
##   4054.7   4125.4  -2014.3   4028.7     1689
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.8253 -0.5435 -0.0076  0.5975  3.1890
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  subj     (Intercept) 0.3285426 0.57319
##  subj.1   TSV_inMC    0.0675844 0.25997
##  subj.2   TSV_inRC    0.0000000 0.00000
##  subj.3   complexity  0.0072340 0.08505
##  item     (Intercept) 0.0008833 0.02972
##  item.1   TSV_inMC    0.0019507 0.04417
##  item.2   TSV_inRC    0.0000000 0.00000
##  item.3   complexity  0.0030094 0.05486
##  Residual             0.5247027 0.72436
## Number of obs: 1702, groups:  subj, 71; item, 24
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept) -2.168053   0.070516 -30.746
## TSV_inMC     0.527612   0.040619  12.989
## TSV_inRC     0.002595   0.024833   0.104
## complexity  -0.280638   0.023143 -12.126
##
## Correlation of Fixed Effects:
##            (Intr) TSV_MC TSV_RC
## TSV_inMC    0.000
## TSV_inRC    0.000  0.000
## complexity  0.000  0.000 -0.001
```

```
m0.MCV2.rrt.nested.2<-lmer(rrt~ MC_inSVO + MC_inTSV + order
         + (1 + MC_inSVO + MC_inTSV + order||subj)
         + (1 + MC_inSVO + MC_inTSV + order||item),
         subset(topic2,constituent =="main-V2"), REML = FALSE)

summary(m0.MCV2.rrt.nested.2) ##  t=-13.89, in SVO, MC faster than RC
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula:
## rrt ~ MC_inSVO + MC_inTSV + order + ((1 | subj) + (0 + MC_inSVO |
##     subj) + (0 + MC_inTSV | subj) + (0 + order | subj)) + ((1 |
##     item) + (0 + MC_inSVO | item) + (0 + MC_inTSV | item) + (0 +
##     order | item))
##    Data: subset(topic2, constituent == "main-V2")
##
##      AIC      BIC   logLik deviance df.resid
##   4049.8   4120.5  -2011.9   4023.8     1689
```

```
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.6940 -0.5384 -0.0019  0.5898  3.2387
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  subj     (Intercept) 0.3288752 0.57348
##  subj.1   MC_inSVO    0.0652057 0.25535
##  subj.2   MC_inTSV    0.0000000 0.00000
##  subj.3   order       0.0141493 0.11895
##  item     (Intercept) 0.0003023 0.01739
##  item.1   MC_inSVO    0.0000000 0.00000
##  item.2   MC_inTSV    0.0014547 0.03814
##  item.3   order       0.0029125 0.05397
##  Residual             0.5192883 0.72062
## Number of obs: 1702, groups:  subj, 71; item, 24
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept) -2.16801    0.07035 -30.815
## MC_inSVO    -0.54291    0.03909 -13.889
## MC_inTSV    -0.01795    0.02592  -0.693
## order       -0.26521    0.02502 -10.602
##
## Correlation of Fixed Effects:
##          (Intr) MC_SVO MC_TSV
## MC_inSVO 0.000
## MC_inTSV 0.000  0.000
## order    0.000  0.000  0.000
```